

R470: Quadratic representation obstruction

TFPT RH programme

31 August 2026

Claim boundary. This note seals one finite obstruction to a named Lean channel map. It does not prove the Gauss–Mellin–digamma identity, compact-window positivity, or the Riemann hypothesis.

1 The named remainder

Round 464 split the selected A_{cap} bridge into a finite PSD half (proved) and the channel `SelectedReadQuadraticRepresentation`:

$$\forall k > 0, \forall f \text{ with } \text{meshExp } f \leq m_k, \exists z \in \mathbb{R}^{\text{cap}+1} \text{ such that } \text{fullRead}(a_k, m_k, f) = z^T A_{\text{cap}} z.$$

The original quantifier does not bound `GridElement.steps`. A degree- $<$ cap identification is therefore not a proof of that proposition. Explicitly out of scope: any statement about infinitely many k (r469 anti-list).

2 Proved arithmetic witness

Lean now proves

`exists\_mesh\_compatible\_steps\_gt\_cap`.

The witness is $k = 5$, $\text{meshExp} = 19$, $\text{steps} = 11$, $\text{cap} = 10$. This is the r464 arithmetic mismatch, now a theorem, not a probe comment.

3 Exact finite obstruction

The channel asked for is stronger than a degree bound: it asks that every mesh-compatible native read lie in the quadratic image of the finite matrix A_{cap} . Even after the honest onset guard $\text{elementAnchor } f \leq a_k$, this fails at $k = 5$.

Take $\text{meshExp} = 3$, $\text{steps} = 24$. Then $\text{support} = 3$, so $\text{elementAnchor} = \lceil e^3 \rceil = 21 \leq a_5 = 32$, and $\text{meshExp} \leq m_5 = 19$. The Hessian T of $\text{fullRead}(a_5, m_5, \cdot)$ on $\mathbb{R}^{24}$ satisfies

$$\text{rank}(T) = 24 > \text{cap} + 1 = 11, \quad \lambda_{\min}(T) = -0.042885425277, \quad \#\{\lambda < 0\} = 2.$$

The selected bordered Hankel A_{cap} at the same window is positive definite ($\lambda_{\min}(A_{\text{cap}}) = 6.61842 \cdot 10^{-8}$, $q^\dagger = 0.877898027321 < 1$). A negative native read cannot equal $z^T A_{\text{cap}} z$ for any z .

The named obstruction (not a `sorry`) is

`SelectedOnsetCompatibleNegativeRead`:

$\exists k > 0, \exists f \text{ with } \text{meshExp } f \leq m_k, \text{elementAnchor } f \leq a_k, \text{ and } \text{fullRead}(a_k, m_k, f) < 0.$

The finite implication

`quadraticRepresentation\_refuted\_of\_negative\_read`

is proved: a negative mesh-compatible read on a PSD cap refutes `SelectedReadQuadraticRepresentation`. The original proposition is retained, unweakened, and unasserted.

4 Kill table

Candidate	Status	Reason
Degree- $<$ cap channel	killed	Proved witness steps $>$ cap at $k = 5$.
Onset guard rescues the map	killed	$k = 5$, meshExp = 3, steps = 24 is onset-compatible and T is indefinite.
Mere existential z on a PSD cap	killed	$\lambda_{\min}(T) < 0 = \lambda_{\min}(A_{\text{cap}})$.
Linear intertwiner $\Phi^T A \Phi = T$	killed	$\text{rank}(T) = 24 > \text{cap} + 1$.
R470 proves RH	killed	One finite obstruction; census stays 8.

5 Census

Lean **sorry** census remains 8: three in **Canonical.lean**, one named Gauss–Mellin–digamma identity in **Elementwise.lean**, three external arrows in **ExternalBridges.lean**, and one source theorem in **Source.lean**. No new **sorry**. Smoke 10/10, SPEC_SHA 74de4e263ac7fd7b.